

The market value of variable renewables

Dispatch and variable generation

These solutions are written to be read after the workshop. Each question is worked in full, in the notation of *Energy System Models*; where the class only covered the key step, the rest is here. The common wrong turns are named where they occur, because they are the useful part.

1 The value of wind in a two-hour system

Part 1 · Mechanism

The system has two hours of equal length, a calm one ($h = 1$, $\gamma_1 = 0$) and a windy one ($h = 2$, $\gamma_2 = 1$), loads $D_1 = 1.5$ and $D_2 = 3$ GW, coal ($\bar{q}_c = 2$, $c_c = 40$), gas ($\bar{q}_g = 2$, $c_g = 80$) and wind ($\bar{q}_w = K$, $c_w = 0$).

1. Prices as a function of K

Nothing links the two hours, so the problem separates into one dispatch problem per hour, and the Lagrangian of the whole is the sum of the hourly ones. In hour h the planner solves

$$\begin{aligned} \min_{q_{w,h}, q_{c,h}, q_{g,h}} \quad & 40 q_{c,h} + 80 q_{g,h} \\ \text{s.t.} \quad & q_{w,h} + q_{c,h} + q_{g,h} = D_h \quad (\lambda_h) \\ & 0 \leq q_{w,h} \leq \gamma_h K, \quad 0 \leq q_{c,h} \leq 2, \quad 0 \leq q_{g,h} \leq 2 \quad (\underline{\mu}_{g,h}, \bar{\mu}_{g,h}). \end{aligned} \quad (1)$$

Written, as in the note, for the maximisation of minus the cost, the Lagrangian is

$$\mathcal{L}_h = - \sum_g c_g q_{g,h} + \lambda_h \left(\sum_g q_{g,h} - D_h \right) + \sum_g \bar{\mu}_{g,h} (\gamma_{g,h} \bar{q}_g - q_{g,h}) + \sum_g \underline{\mu}_{g,h} q_{g,h}, \quad (2)$$

and the first-order condition for each plant is

$$\lambda_h = c_g + \bar{\mu}_{g,h} - \underline{\mu}_{g,h}, \quad \bar{\mu}_{g,h} (\gamma_{g,h} \bar{q}_g - q_{g,h}) = 0, \quad \underline{\mu}_{g,h} q_{g,h} = 0. \quad (3)$$

The reading of (3) is the merit order (Result 1). A plant strictly between its bounds has both multipliers zero and therefore $\lambda_h = c_g$: it sets the price. A plant at its upper bound has $\bar{\mu}_{g,h} = \lambda_h - c_g \geq 0$, so it is cheaper than the price and earns that rent. A plant at zero has $\underline{\mu}_{g,h} = c_g - \lambda_h \geq 0$, so it is dearer than the price and idle. The price is always the marginal cost of the plant that is *interior*.

The calm hour. Wind offers nothing, coal serves all 1.5 GW with 0.5 GW to spare, gas is idle. Coal is interior for every K , so

$$\lambda_1 = 40 \quad \text{for all } K \geq 0. \quad (4)$$

The windy hour. Wind is cheapest, so it runs first and the thermal plants face the residual load $R_2 = 3 - K$. Which plant is interior depends on where R_2 falls:

	$R_2 = 3 - K$	dispatch (wind, coal, gas)	interior plant	λ_2
$0 \leq K < 1$	$(2, 3]$	$K, 2, 1 - K$	gas	80
$1 < K < 3$	$(0, 2)$	$K, 3 - K, 0$	coal	40
$K > 3$	< 0	$3, 0, 0$	wind, curtailed by $K - 3$	0

Figure 1 draws the same thing as a merit order: the supply staircase of the windy hour, with wind's block widening as K grows, and the load fixed at 3 GW. The price is the height of the step the load line cuts.

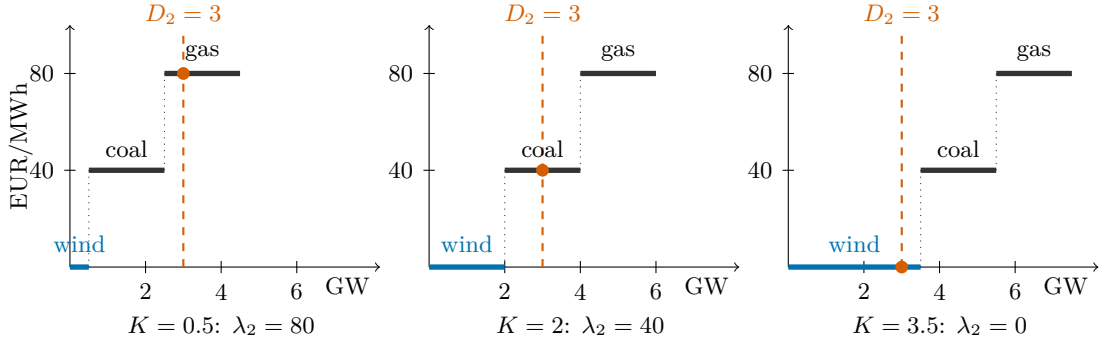


Figure 1: The merit order of the windy hour at three levels of wind capacity. Wind's block (zero cost) pushes coal and gas to the right; the load line at 3 GW cuts the staircase at gas, at coal, and finally at wind itself, which is then curtailed.

At $K = 1$ exactly, coal is at its bound and gas at zero; neither is interior and any $\lambda_2 \in [40, 80]$ satisfies (3). At $K = 3$ the same happens between coal and wind, with $\lambda_2 \in [0, 40]$. These two values of K are the kinks: the price-setting plant changes there, and nowhere else.

Remark. The common wrong turn is at $K > 3$: to keep coal “marginal” and the price at 40. But coal is off, with $\mu_{c,2} = 40 - \lambda_2 > 0$, and an idle plant does not set the price. Wind is the plant strictly between its bounds, $0 < q_{w,2} = 3 < K$, so (3) for wind gives $\lambda_2 = c_w = 0$. Curtailment is not a malfunction; it is the price telling the system that the last MWh of wind is worth nothing in that hour.

2. Capture price, base price and value factor

The three definitions are

$$\bar{p}_w = \frac{\sum_h \lambda_h q_{w,h}}{\sum_h q_{w,h}}, \quad \bar{p} = \frac{\sum_h \lambda_h D_h}{\sum_h D_h}, \quad VF_w = \frac{\bar{p}_w}{\bar{p}}. \quad (5)$$

Wind produces only in the windy hour, so the capture price is that hour's price whatever the amount produced:

$$\bar{p}_w(K) = \frac{\lambda_2 q_{w,2}}{q_{w,2}} = \lambda_2(K). \quad (6)$$

The base price weights the two prices by their loads,

$$\bar{p}(K) = \frac{40 \cdot 1.5 + \lambda_2(K) \cdot 3}{4.5} = \frac{40}{3} + \frac{2}{3} \lambda_2(K). \quad (7)$$

Putting in the three values of λ_2 :

	$\bar{p}_w$	$\bar{p}$	VF_w
$0 \leq K < 1$	80	66.7	1.2
$1 < K < 3$	40	40	1.0
$K > 3$	0	13.3	0

$VF_w(K)$ is a decreasing step function with two drops. At $K = 1$ the last MW of gas leaves the windy hour and the price there falls from 80 to 40. At $K = 3$ the last MW of coal leaves, wind is curtailed, and the price falls to zero. Between the kinks *nothing happens*: adding wind from $K = 1.5$ to $K = 2.5$ changes coal's output and nothing else.

Two features of this answer carry over to the general case. The base price falls too, from 66.7 to 40 to 13.3, because the windy hour is two-thirds of the load; but it falls less than the capture price, because the calm hour's 40 is in the base price and not in wind's. And the whole decline is a sequence of discrete events in the merit order, not a smooth curve; the smooth curves of part 2 are thousands of such events at different hours.

3. Hirth's base price against the note's

Hirth (2013, §3.1) defines the base price as the time-weighted average, $\langle \lambda \rangle = \frac{1}{T} \sum_h \lambda_h$. At $K = 0.5$ that is $(40 + 80)/2 = 60$, so his value factor is

$$VF_w^H = \frac{80}{60} = 1.33 \quad \text{against the note's} \quad VF_w = \frac{80}{66.7} = 1.2. \quad (8)$$

For the general condition, write $\bar{D} = \frac{1}{T} \sum_h D_h$ and subtract the two base prices:

$$\bar{p} - \langle \lambda \rangle = \frac{\sum_h D_h \lambda_h}{T \bar{D}} - \frac{\sum_h \bar{D} \lambda_h}{T \bar{D}} = \frac{1}{T \bar{D}} \sum_h (D_h - \bar{D}) \lambda_h = \frac{\text{Cov}(D, \lambda)}{\bar{D}}. \quad (9)$$

The last step is where the level of the price disappears, and it deserves a line: the covariance is $\frac{1}{T} \sum_h (D_h - \bar{D})(\lambda_h - \langle \lambda \rangle)$, and the term in $\langle \lambda \rangle$ is $\langle \lambda \rangle \cdot \frac{1}{T} \sum_h (D_h - \bar{D}) = 0$, because

deviations from a mean sum to zero. So $\sum_h (D_h - \bar{D})\lambda_h$ and $\sum_h (D_h - \bar{D})(\lambda_h - \langle\lambda\rangle)$ are the same number, and only the *co-movement* of load and price survives. The consumption-weighted base price exceeds the time average exactly when prices and loads are positively correlated across hours, and then Hirth’s value factor, which divides by the smaller number, is the larger. That is the normal case in every real system, because load is what pushes the price up the merit order. The consumption-weighted definition is therefore the more demanding benchmark: it compares wind with what the average MWh *consumed* costs, not with the average hour. The difference matters most for solar, whose output coincides with the midday load peak; in part 2’s table the two bases differ by about one percent for this fleet, because at every eighth hour the price–load correlation is modest.

4. Short term, and the capture price as a marginal cost saving

Hirth (2013, §4.3) sorts models by how they treat the capital stock: *short term* takes the fleet as given and solves dispatch only; *mid term* adds investment and retirement on top of the existing fleet; *long term* builds the whole system from scratch. This exercise is his short term: K is a parameter, the thermal fleet never changes, and the value of wind is what Hirth (2013) following Grubb calls its “marginal fuel-saving value”. Only in the long term are all profits zero (Result 6); in the short term the inframarginal plants earn rents.

Now the derivative. Let $C^*(K)$ be the minimised system cost of the general T -hour problem. The only place K enters is wind’s capacity constraint, whose Lagrangian term is

$$\sum_h \bar{\mu}_{w,h} (\gamma_h K - q_{w,h}). \quad (10)$$

The envelope theorem says that the derivative of the optimal value with respect to a parameter is the partial derivative of the Lagrangian at the optimum. Since the Lagrangian is written for $-C$,

$$\frac{\partial(-C^*)}{\partial K} = \sum_h \bar{\mu}_{w,h} \gamma_h \quad \implies \quad \frac{\partial C^*}{\partial K} = - \sum_h \gamma_h \bar{\mu}_{w,h}. \quad (11)$$

In an hour where wind is not curtailed it is at its upper bound, so $\underline{\mu}_{w,h} = 0$ and (3) gives $\bar{\mu}_{w,h} = \lambda_h - c_w = \lambda_h$. Hence

$$\frac{\partial C^*}{\partial K} = - \sum_h \gamma_h \lambda_h. \quad (12)$$

If no hour is curtailed, one MW of wind produces $\sum_h \gamma_h$ MWh over the year, so the saving *per MWh of wind* is

$$\frac{\sum_h \gamma_h \lambda_h}{\sum_h \gamma_h} = \bar{p}_w, \quad (13)$$

which is the capture price of (5) once $q_{w,h} = \gamma_h K$ is substituted. This is Hirth’s statement that “the market value of VRE equals marginal cost savings in the power system”, derived: what a wind MWh earns in the market is what it saves the planner. In the two-hour system,

(12) gives -80 , -40 and 0 per MW on the three intervals, and the zero is worth noting: in a curtailed hour $\bar{\mu}_{w,h} = 0$, and an extra MW of wind saves nothing because it produces nothing that is used.

Remark. The qualifier “wherever wind is not curtailed” in the question is doing work, and it is worth seeing exactly what breaks without it. In a curtailed hour wind is interior, so $\bar{\mu}_{w,h} = 0$ and $\lambda_h = c_w = 0$: both sides of (12) lose that hour, and the derivative $\partial C^*/\partial K = -\sum_h \gamma_h \lambda_h$ still holds. What fails is the sentence after it. One MW no longer produces $\sum_h \gamma_h$ MWh but only $\Gamma(K) \equiv \sum_h q_{w,h}/K \leq \sum_h \gamma_h$, the *realised* full-load hours: the full γ_h in the uncurtailed hours plus whatever the load absorbs in the curtailed ones, a number that falls with K . The capture price is the saving per MWh *delivered*,

$$\bar{p}_w = \frac{\sum_h \lambda_h q_{w,h}}{\sum_h q_{w,h}} = \frac{\sum_h \gamma_h \lambda_h}{\Gamma(K)} \geq \frac{\sum_h \gamma_h \lambda_h}{\sum_h \gamma_h},$$

and it is the first equality that is Hirth’s statement in general: what a delivered MWh earns is what it saves. Dividing the saving by the *potential* output instead understates the capture price, by the curtailed share. In the two-hour system at $K = 3.5$ both are zero, so nothing shows; in part 2 the gap opens as soon as curtailment does.

5. Monotonicity, and the sign at zero share

Substituting $q_{w,h} = \gamma_h K$ into the capture price,

$$\bar{p}_w = \frac{\sum_h \lambda_h \gamma_h K}{\sum_h \gamma_h K} = \frac{\sum_h \gamma_h \lambda_h}{\sum_h \gamma_h}. \quad (14)$$

K has cancelled. Wind capacity affects wind’s capture price *only through the prices*, and (14) is a weighted mean of the prices with fixed weights γ_h . So it is enough to show that every λ_h is non-increasing in K . In hour h one more MW of wind lowers the residual load by γ_h ; the price is the marginal cost of the plant that is interior at that residual load; and as residual load falls, the interior plant moves down the merit order or stays put (question 1 showed this on two hours). So $\lambda_h(K)$ is a non-increasing staircase in K , and a weighted mean of non-increasing functions is non-increasing.

Remark. The substitution $q_{w,h} = \gamma_h K$ is again the no-curtailment case, and here the qualifier is not cosmetic. With partial curtailment the delivered MWh in a curtailed hour are fixed by the load and sell at zero, so between two kinks the capture price per delivered MWh is $K \sum_u \gamma_u \lambda_u / (K \sum_u \gamma_u + Q_c)$, with u the uncurtailed hours and Q_c the energy delivered in the curtailed ones, and that *rises* in K : the zero-price energy is a shrinking share of what wind sells. Two hours with $\gamma = 1$ in both and loads 1.5 and 0.5 show it: for $0.5 < K < 1.5$ the first hour pays coal’s 40 and the second is curtailed at zero, so $\bar{p}_w = 40K/(K + 0.5)$, from 20 at $K = 0.5$ to 30 at $K = 1.5$. Monotonicity is a property of the uncurtailed capture price, which is the object question 4 and part 3 work with; the sweeps of part 2 fall throughout, but that is the profiles, not a theorem.

For the sign at zero share, the intuition first: **the value factor is above one when wind blows in the expensive hours**, that is, when wind's availability is positively correlated with the price. This is Hirth's correlation effect, and since the price without wind is a non-decreasing function of the load, it is the correlation of wind with load. In the two-hour system the windy hour is the high-load hour, priced by gas, and $VF_w = 1.2$; swap the two loads and the windy hour is priced by coal while the calm hour needs gas, giving $\bar{p}_w = 40$, $\bar{p} = 66.7$ and $VF_w = 0.6$. Same wind, same fleet, other side of one.

Now the exact statement, and where it departs from the intuition. Let λ_h^0 be the prices without wind, to which the prices tend as $K \rightarrow 0$, and let $\langle \lambda^0 \rangle$ be their time average. The capture price is a γ -weighted mean of the prices, so by the same step as (9),

$$\bar{p}_w - \langle \lambda^0 \rangle = \frac{\text{Cov}(\gamma, \lambda^0)}{\bar{\gamma}}, \quad \bar{p} - \langle \lambda^0 \rangle = \frac{\text{Cov}(D, \lambda^0)}{\bar{D}}, \quad (15)$$

with $\bar{\gamma}$ and $\bar{D}$ the time averages (the level $\langle \lambda^0 \rangle$ drops out of both covariances for the reason given under (9)). Hence

$$VF_w > 1 \iff \frac{\text{Cov}(\gamma, \lambda^0)}{\bar{\gamma}} > \frac{\text{Cov}(D, \lambda^0)}{\bar{D}}. \quad (16)$$

With Hirth's time-weighted base price the right-hand side is zero and the condition is exactly the intuition: $VF_w > 1$ if and only if wind is positively correlated with the price. With the consumption-weighted base price the benchmark is not a flat profile but the load's own profile, which is itself positively correlated with the price. So the exact condition reads: wind must be *more* positively correlated with the price, per unit of its mean, than the load is. That is a higher bar, and it is why the same wind can have a value factor above one on Hirth's definition and below one on the note's (question 3).

Stop point

$VF_w(K) = 1.2$ for $K < 1$, 1.0 for $1 < K < 3$, 0 for $K > 3$; the kinks are the last MW of gas leaving at $K = 1$ and the last MW of coal leaving (curtailment starting) at $K = 3$. And $\partial C^*/\partial K = -\sum_h \gamma_h \lambda_h$.

Stretch question

Is VF_w necessarily non-increasing in K ? No. The argument above shows that both the numerator and the denominator of the value factor fall; it does not show which falls faster. Build a system in which the price falls in an hour that carries little weight in wind's average and much in the load's. Hour A: $\gamma_A = 1$, $D_A = 1.5$, so coal is interior and $\lambda_A = 40$ for all small K . Hour B: $\gamma_B = 0.1$, $D_B = 2 + \varepsilon$, so gas is interior with $\lambda_B = 80$ until $0.1K > \varepsilon$ pushes gas out, after which $\lambda_B = 40$. Across that K ,

$$\begin{aligned}\bar{p}_w &: \frac{1 \cdot 40 + 0.1 \cdot 80}{1.1} = 43.6 \rightarrow 40, \\ \bar{p} &: \frac{1.5 \cdot 40 + 2 \cdot 80}{3.5} = 62.9 \rightarrow 40, \\ VF_w &: 0.69 \rightarrow 1.0.\end{aligned}$$

The value factor *rose*. In a real year the price falls mostly in the hours where the wind is, because that is where wind cuts residual load most, and the numerator falls faster: the sweeps of part 2 show it. But that is a property of the profiles, not a theorem.

2 The cannibalisation curve, and what drives it

Part 2 · Evidence

The evidence is the hourly dispatch model on the ten-plant fleet, DK1 2024 profiles at every eighth hour. Marginal costs, in EUR/MWh: solar 0, offshore wind 0, onshore wind 1.5, waste 28, coal 37, CCGT 66, OCGT 91, wood chips 98, wood pellets 146, oil 163. Keep that ladder in view; it is the staircase every answer below walks.

1. Slopes against Hirth's regression

From the wind block of the worksheet's table, with the value factor in percentage points so that the units match Hirth's,

$$\frac{87 - 101}{33 - 3} = -0.47 \quad \text{and} \quad \frac{61 - 87}{59 - 33} = -1.0 \quad (17)$$

percentage points of the value factor per percentage point of share. Hirth's regression (his eq. 4, Table 5) gives -1.62 in thermal systems and -0.26 in hydro systems, from an intercept near one (his text quotes -0.22 for hydro; the table says -0.26). Two reasons for the differences, pulling in opposite directions, and of different kinds: the first is a difference between the two *systems* and would be there whatever the method; the second is a difference between the two *methods*, a regression across years against a model that holds the fleet fixed.

The slope of the merit order. Hirth’s Table 8 names the statistic that decides the size of the drop: how far the price falls for each GW of residual load that wind removes. That is the slope of the merit-order curve, and the two fleets differ in it. Reading each ladder from the top down, in the order wind pushes the price through it:

Hirth’s thermal Europe	this fleet (Table 1, capacities in GW)
gas turbines	OCGT 91 (0.8), wood 98–146 (0.95)
CCGT	CCGT 66 (1.5)
hard coal	coal 37 (1.2)
lignite	waste 28 (0.35)
nuclear	—
zero (wind itself)	zero (wind itself)

Below the CCGT block the two ladders part company. In Hirth’s system the price keeps stepping down, from hard coal to lignite to nuclear, each with its own block of capacity, before it reaches zero: the descent is spread over many GW and every GW of wind takes another step. Here the price drops once, from 66 to 37, and then rests on coal’s plateau for 1.2 GW and waste’s for another 0.35 GW before it falls to zero. So across the range where the value factor is measured, a GW of wind moves the price less than in his systems. That is what a flatter merit order means: not that prices are low, but that wind moves them little until the cliff at zero, which is the second regime of question 2.

A second difference acts at the top of the curve rather than along it. Danish wind is positively correlated with Danish load (+0.44 hour by hour, Table 2), so wind is paid in above-average hours to begin with, and the curve starts higher. Both differences make the model’s slope smaller than Hirth’s thermal estimate.

What adjusts. A decade of market data is not a fixed fleet. Between 2001 and 2010 plants were built and retired around the growing wind fleet, which is Hirth’s mid-term and long-term adjustment; in his own model it raises the value factor at 30% from 0.5 to 0.65. A regression across years therefore sees a flatter decline than a short-term model sweeping one fixed fleet would. Two further reasons to read his coefficients loosely: the shares in his sample are mostly below 20%, on the flat part of the curve, and the identification is across countries and years with time dummies, not a sweep of one system.

The fair comparison is therefore the model’s second interval, -1.0 against -1.62 : same order of magnitude, smaller because the staircase is flatter. The first interval is the flat top that every curve has.

2. Two regimes, one mechanism

Below $s \approx 1.5$ no energy is curtailed, yet the value factor falls from 0.97 to 0.87. The mechanism is the one of part 1 at $K = 1$: in the windy hours wind pushes the residual load down the staircase, so hours that were priced by the OCGT or by wood become

CCGT hours, and CCGT hours become coal hours. Every such step lowers the price in a windy hour, and windy hours are where wind is paid. This is the merit-order effect between plants; Hirth's Fig. 2 draws it.

Beyond $s = 2$ the residual load turns negative in a growing number of hours. There wind is the plant strictly between its bounds, so the price is its own marginal cost, zero (or onshore's 1.5), and the energy above the load is curtailed: 9% of the energy offered at $s = 2$, 43% at $s = 4$, 59% at $s = 6$. This is part 1 at $K = 3$.

What curtailment does to the quantities. The value factor and the capture price are prices, and in the second regime they miss half the story. A curtailed MWh is not sold at zero; it is not sold at all. So the energy a MW of wind delivers over the year, its realised capacity factor, falls with s as well, and revenue per MW is the product of a falling price and a falling quantity. From the sweep, with the delivered energy per MW read off the table's revenue and capture price:

s	curtailed %	capacity factor	$\bar{p}_w$	revenue per MW, $s = 1$ as 100
1	0	0.36	49.2	100
1.5	1	0.36	35.7	72
2	9	0.33	22.5	41
3	29	0.26	13.5	19
4	43	0.21	10.1	12
6	59	0.15	6.9	6

In the first regime the capacity factor is untouched at 0.36, the profiles' own value, and every loss of revenue is a loss of price. From $s = 1$ to $s = 4$ the capture price falls by a factor of 4.9 and the capacity factor by a factor of 1.7, so revenue per MW falls by a factor of 8.5: the quantity channel adds seventy percent on top of the price collapse. Two consequences. For the investor, the levelised cost of a MW is its annualised cost over the MWh it *delivers*, so curtailment raises the LCOE of every farm at the same time as it lowers the price each delivered MWh earns; part 3, question 1 uses this. For the reader of the value factor, the ratio is silent about all of it: a MWh that is never produced enters neither the numerator nor the denominator, which is one more reason to report capture prices and revenue in levels, as question 4 will insist.

The two regimes are one mechanism because the rule is the same in both: the price in an hour is the marginal cost of the interior plant, and wind walks that staircase down until it reaches its own step. What changes between the regimes is only which step is being taken: from one thermal plant to the next, or from the cheapest thermal plant to wind itself. Hirth's Fig. 11 shows the price-setting technology by share of hours, and the share of wind energy sold at zero price. For this fleet at $s = 4$ it would show wind price-setting in the majority of hours, coal and the CCGT in the rest, and well over half of the energy wind sells sold at a zero price, with 43% of what the weather offered not sold at all.

3. Why solar falls faster

From the solar block, over its first fifteen points of share and in the same units as question 1,

$$\frac{74 - 106}{15 - 0.2} = -2.2 \quad (18)$$

percentage points per percentage point, against wind’s -0.47 over its first thirty: about five times faster. The reason is concentration. Solar’s capacity factor is 0.13 against 0.30 for onshore and 0.45 for offshore wind, so a given share of *energy* is delivered in a third as many hours. And every solar plant delivers it in the same hours. The statistic that measures this is the correlation of solar with load across the hours of the day, $+0.94$: solar output is a single midday rhythm. A share of x percent of consumption is therefore concentrated into the midday hours, and in those hours it pushes the price down the whole staircase at once. Wind’s share is spread over the day and the year, so the same share lands more thinly and the price steps are smaller and fewer. Hirth’s conclusion says exactly this: solar reaches wind’s value drop at half the share “because its generation is concentrated in fewer hours”.

4. The decomposition of the ratio trap

In the third sweep solar is scaled while wind is held fixed. Between $s = 1$ and $s = 24$:

	$s = 1$	$s = 24$	$\Delta \ln$
wind capture price $\bar{p}_w$	49.2	43.5	-0.12
base price $\bar{p}$	56.8	35.3	-0.48
wind value factor VF_w	0.87	1.23	$+0.35$
wind revenue, MEUR	49	33	-0.39

Since $VF_w = \bar{p}_w / \bar{p}$,

$$\Delta \ln VF_w = \Delta \ln \bar{p}_w - \Delta \ln \bar{p} = -0.12 - (-0.48) = +0.35. \quad (19)$$

Wind’s value factor rose by 35 log points, not because wind earned more, but because the base price fell by 48 log points while wind’s capture price fell by 12. Solar collapsed the midday prices, which are a large part of the base price and a small part of what wind earns.

In the vocabulary of López Prol et al. (2020), the capture-price term is *absolute* cannibalisation, here the cross effect of solar on wind: wind’s revenue per MWh fell. The value-factor term is *relative* cannibalisation, wind’s value relative to the average MWh, and it has the opposite sign. A wind investor reads the capture price and the revenue, and both fell; revenue by more than the capture price, because at high solar shares some wind is curtailed too. A policy maker who indexes support to the value factor reads a number

that improved while everything the investor cares about got worse. This is the ratio trap, and it is why Prol et al. insist on reporting both.

Stop point

Wind: -0.47 and -1.0 points per percentage point. Solar: about five times wind's early slope. Decomposition: $\Delta \ln \bar{p}_w = -0.12$, $\Delta \ln \bar{p} = -0.48$.

Stretch question

The carbon price. Hirth (2013, §5.5) names three channels: a higher carbon price raises the price level; it flattens the merit order between lignite, coal and gas; and it induces investment in low-carbon dispatchable plant (nuclear, CCS in his model) with low variable cost, which steepens it. The third does not exist here, because the fleet is fixed. The second does: at $\tau = 85$ EUR/t coal goes from 37 to 117 and the CCGT from 66 to 96, so the middle of the merit order becomes CCGT (96), wood chips (98), coal (117), with gaps of 2 and 19, where before it was coal (37), CCGT (66) with a gap of 29. Flatter. The first channel also acts on the *ratio*, which is easy to miss: the tax adds roughly the same amount to every thermal price, and adding a constant a to a numerator and a denominator with ratio below one, $(x + a)/(y + a)$ with $x < y$, moves the ratio towards one, while the surplus hours stay at a price of zero. Both channels push wind's value factor up, so the prediction is: up. The notebook confirms it, $0.61 \rightarrow 0.74$ at $s = 2$ and $0.87 \rightarrow 0.97$ at $s = 1$.

3 Optimal deployment, and whether the drop is a market failure

Part 3 · Critique

1. Hirth's figure 3 from the model

Let wind cost c^I per MW and year, annualised, and let $\Gamma = \sum_h \gamma_h$ be the full-load hours of one MW. From part 1, one MW of wind earns

$$\sum_h \gamma_h \lambda_h(K) = \Gamma \bar{p}_w(K) \quad (20)$$

per year. Under free entry, farms are built as long as this exceeds c^I , and entry stops where

$$\Gamma \bar{p}_w(K) = c^I \quad \Longleftrightarrow \quad \bar{p}_w(K) = \frac{c^I}{\Gamma} \equiv LCOE_w. \quad (21)$$

The right-hand side is wind's levelised cost, its annualised cost spread over the MWh it produces, and (21) says entry stops where the market value of a wind MWh equals its levelised cost: Hirth's intersection. His falling curve is $\bar{p}_w(K)$, non-increasing by part 1,

question 5. His rising curve is a resource constraint, the next site being worse than the last, which this model does not have; here the LCOE is flat and the intersection is still unique because the other curve falls.

Remark. $\Gamma = \sum_h \gamma_h$ is the full-load hours of an uncurtailed MW, as in part 1, question 4. Once wind is curtailed the revenue of one MW is still $\sum_h \lambda_h q_{w,h}/K = \sum_h \gamma_h \lambda_h$, because the curtailed hours carry a zero price, so the free-entry condition in its first form, $\sum_h \gamma_h \lambda_h(K) = c^I$, and the planner's condition below are unaffected. What changes is the translation into levelised cost: the MWh actually delivered per MW are the realised full-load hours $\Gamma(K) \leq \Gamma$ of the remark in part 1, so the condition reads $\bar{p}_w(K) = c^I/\Gamma(K)$, and the levelised cost per delivered MWh *rises* with K . That gives Hirth's rising curve a second cause besides site quality: at high shares each MW delivers less, so each delivered MWh must carry more of the capital cost.

Now the planner. She chooses K to minimise total cost, dispatch plus capital,

$$\min_K C^*(K) + c^I K \quad \implies \quad -\frac{\partial C^*}{\partial K} = c^I. \quad (22)$$

By part 1, question 4, the left-hand side is $\sum_h \gamma_h \lambda_h = \Gamma \bar{p}_w(K)$, so the planner's condition is (21): the same equation. This is Result 6, that a technology built at the optimum earns exactly its annualised cost in scarcity rents, and it is the first welfare theorem for the capacity margin: the competitive market builds the planner's amount of wind.

With a production subsidy σ per MWh, a farm earns $\Gamma(\bar{p}_w + \sigma)$ and entry continues until

$$\bar{p}_w(K) + \sigma = LCOE_w, \quad (23)$$

which, since $\bar{p}_w$ is non-increasing, is at a larger $K^\sigma > K^*$ wherever the curve is strictly falling. Every MW between K^* and K^σ costs c^I and saves $\Gamma \bar{p}_w(K) < c^I$ in dispatch cost; the welfare loss is

$$\int_{K^*}^{K^\sigma} (c^I - \Gamma \bar{p}_w(K)) dK, \quad (24)$$

the area between the LCOE line and the market-value curve to the right of Hirth's intersection. The subsidy is a transfer; the area is the loss.

Stop point

Free entry stops at $\Gamma \bar{p}_w(K) = c^I$, that is $\bar{p}_w = LCOE_w$, and this is also the planner's condition (Result 6).

2. Placing the developer's argument

The developer describes the falling capture price: Hirth's *profile cost*. It travels entirely through the price, and the price is doing its job. Electricity in windy hours has genuinely become less scarce; the hourly price is the marginal cost of serving the last MWh (Result

1), and a lower price in windy hours is the correct signal to a battery, a heat pump or an electrolyser to consume then. Nothing happens outside the price system, so it is a pecuniary externality and not a market failure; and by question 1, “compensating” it pushes past the optimum. The bakery is the right comparison: a new bakery lowers the price of bread and harms the incumbent baker, and the allocation is better for it. Compensating the incumbent would make it worse.

Hirth’s other two costs are where §3.5 of the note goes further than his sentence, and it does so conditionally.

- *Grid-related costs* are a technological externality when the farm aggravates congestion that it does not pay for, because tariffs are flat or because zonal pricing hides the constraint. The targeting principle prescribes pricing the congestion: nodal pricing, or locational connection charges.
- *Balancing costs* are a technological externality when the reserve the system operator holds against the farm’s forecast error is procured centrally and socialised through a flat tariff. The instrument is an imbalance charge on whoever causes the imbalance.

In both cases the condition is “unpriced”, and in both the remedy is one instrument per externality, applied at the source. Neither is a reason to pay wind per MWh, and neither has anything to do with profile costs.

Remark. The common wrong turn is to add price *volatility* to the list of market failures. Volatility is unpleasant for an investor, but it is risk, and risk can be carried by contracts: fixed-price power purchase agreements, hedges, contracts for difference. It becomes an argument for intervention only where those contracts are missing, which is a failure in a different market and the subject of lecture 7.

3. Joskow, in the language of the model

The good is electricity delivered in a particular hour at a particular place. Its price is that hour’s λ_h , and λ_h differs across hours, so a MWh delivered in a windy hour and a MWh delivered in a tight hour are different goods with different prices. A levelised cost is a cost per MWh averaged over a plant’s own output profile. Comparing two LCOEs therefore compares the cost of two different bundles of goods as if they were one good, and ranks the bundles by cost without asking what either is worth.

The two-hour system makes it concrete. At $K = 0.5$ a MWh of wind sells for 80, because wind delivers in the gas-priced hour, while a MWh of coal sells for $(40 \cdot 1.5 + 80 \cdot 2) / 3.5 = 57$ on average over its two hours. At $K = 3.5$ a MWh of wind sells for 0 and a MWh of coal for 40. The same wind and the same coal, and the ranking of their values has reversed; and neither 80, 57, 0 nor 40 appears in an LCOE. The right comparison is the cost of a technology against the market value of what it actually delivers, which is Hirth’s figure,

and question 1 showed that the market value of wind depends on how much wind is already there.

Stretch question

Solar raises wind's value factor in California. Ask for wind's capture price, or its revenue, in levels. Part 2's third sweep is the same experiment in the model: solar entry raised wind's value factor from 0.87 to 1.23 while wind's capture price fell by 12% and its revenue by a third. A ratio that improves while both of its parts get worse is not complementarity; it is a denominator falling faster than a numerator. Prol et al. find the same sign pattern in the Californian data, and report absolute and relative cannibalisation side by side for exactly this reason: the value factor alone would have told the lobbyist's story.

References

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